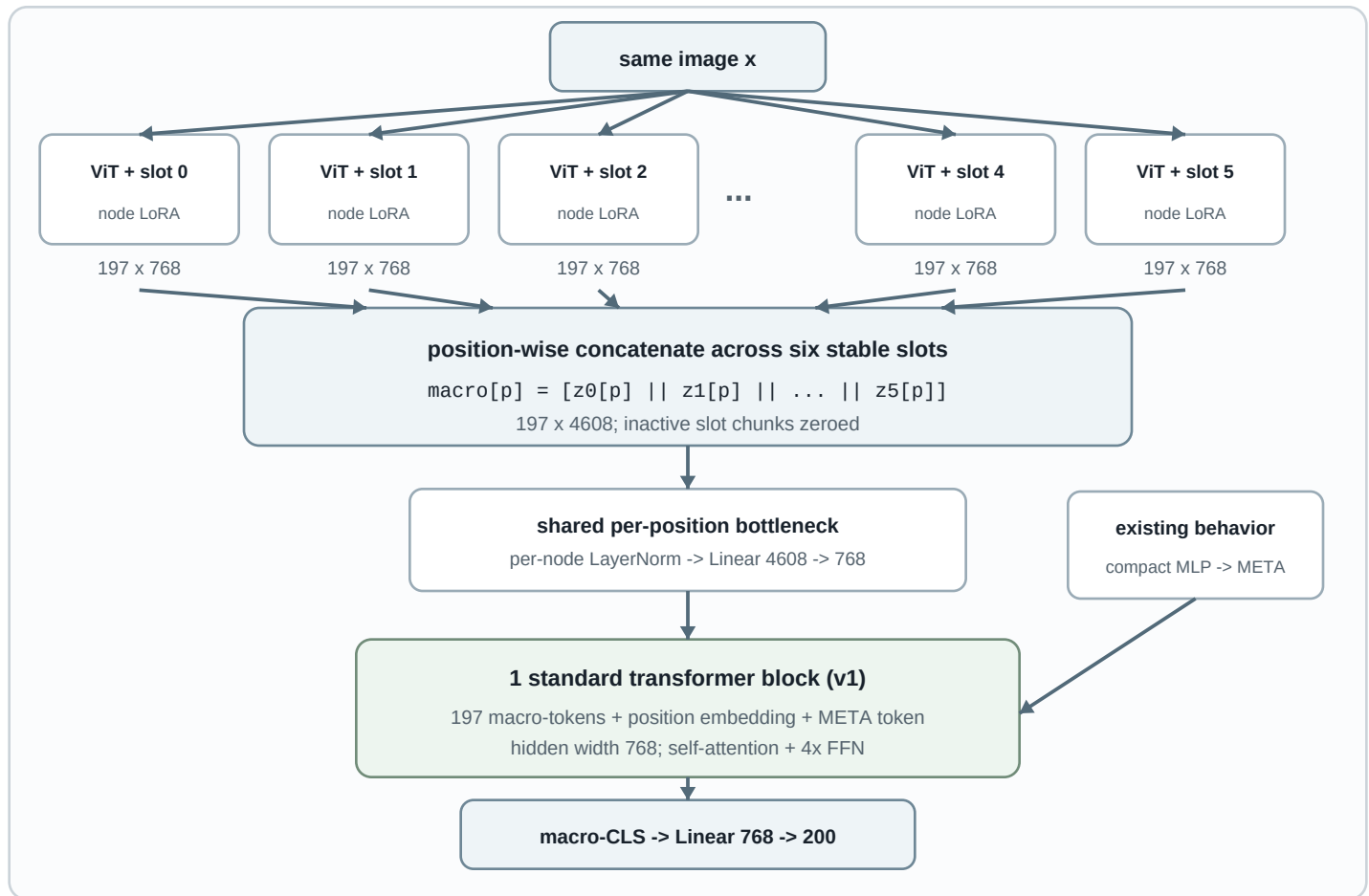


Macro-Token Transformer Integrator

Short reference specification for experiment planning. Status: adopted v1 recommendation.

Core idea. Preserve every node's final 197-token ViT representation. At each position, concatenate the corresponding 768-D token from all six hierarchy slots into one 4608-D macro-token. Compress to 768 dimensions with one shared projection, then let a small transformer integrate spatially. Classify from macro-CLS.

Architecture



Capture point

Use the same final pre-classifier representation point as the current v6 class-token integrator, but retain the full 197-token sequence for each node.

Slot semantics

Keep six hierarchy levels as stable slots. Zero inactive 768-D chunks and preserve an explicit active-slot signal in META. Do not repack live nodes.

Why this is the preferred next integrator

- Restores information discarded by CLS pooling: local patch evidence and node-to-node disagreement remain visible before global compression.
- Uses the natural correspondence already present in the models: position p in every node refers to the same ViT patch position.
- Shares weights across all 197 positions: one 4608->768 cross-node comparison rule is reused everywhere.
- Does not require perfect latent alignment across nodes: every hierarchy slot has its own slice of the input projection.
- Retains existing behavior features: raw scores, local log-probabilities, ownership features, and active bits become a compact META token.

Parameter and data scale

Head	Approx. params	Interpretation
Current v6 MLP	9.12M	Final CLS + behavior
Two-CLS-token MLP	13.84M	Latest latent ablation
Macro-token v1, D=768, 1 block	~10.8M core	3.54M bottleneck + ~7.1M block + classifier
Compact META encoder	~1.1M	Example: 3606->256->768
Macro-token v1 total with META	~11.9M	Baseline target
Macro-token, D=768, 2 blocks	~17.9M core	First capacity ablation
Raw D=4608 transformer, 1 block	~256M	Reject: unnecessarily large

D = hidden width of the new integrator transformer; counts assume a conventional 4x feed-forward expansion.

Do not flatten all node tokens into a giant MLP.

Six nodes x 197 positions x 768 dimensions = 907,776 input features. A single 907,776->768 dense layer needs about 697M weights; 907,776->1024 needs about 930M. The proposed shared bottleneck needs only about 3.54M weights.

Experimental motivation

The latest two-layer latent ablation added a second 768-D global CLS token per node, growing the MLP from 9.12M to 13.84M parameters. Mean locked-test accuracy improved only 0.524 point across matched online cells, while fresh full-history test accuracy changed by -0.079 and -0.167 point at stages 31 and 50. The next test should therefore change the information boundary and integration structure, not merely add another global summary. Current clean development uses 19,200 fitting identities and 4,800 validation identities; the locked benchmark uses 24,000 training and 6,000 test identities. A roughly 12M-parameter head with strong position-wise sharing is plausible. A 0.7B-1B dense union is not.

V1 build contract

BASELINE: D=768 | 1 block | 197 macro-tokens + META | classifier from macro-CLS

- Extract each node full final adapted sequence: 197 x 768.
- Normalize each node token chunk independently over its 768 features.
- Concatenate corresponding positions across six fixed level slots: 197 x 4608.
- Apply one shared Linear(4608, 768); add learned 197-position embedding.
- Encode current nonlatent behavior features through a compact MLP into one 768-D META token.
- Run one 768-wide transformer block; classify with Linear(768, 200).
- First structural ablation: the identical model with two transformer blocks.

Planning guardrails

- Evaluate fresh full-history training first to ask whether the representation raises the integration ceiling before mixing in online replay limitations.
- Keep the hierarchy and source-node checkpoints fixed for the first comparison; change only the integrator representation/head.
- Use an end-to-end clean validation partition for architecture selection; the latest run found that current integrator validation identities had been seen during upstream node fitting.
- Add an owner-prediction probe on the macro representation to separate missing routing information from failure to exploit available routing information.
- Report parameter count, token-cache size, training wall time, and inference cost with accuracy.